

Radial Space — What We Have Discovered

All classifiers closed-form; every number reproducible from radial_map.py / radial_baseline.py / radial_data/*.json

GENREG lab, 2026-07-13 · branch radial-cifar-nogradient · no gradients anywhere

The idea

Characterize activation functions by how they TRANSFORM data (behavior, not formula).
Give every function a deterministic integer address: lens i's program (1-3 composed primitives $\text{prim}(a*x+b)$ from a 14-op catalog) is generated from i alone. Fingerprint each lens on a data stream, project signature distances to 3D (classical MDS) - that is the map.

The five headline discoveries

1. The map is real: linear core at the origin, primitive families in ordered regions, oscillators in their own lobe. Determinism error 0.0 in every domain.
2. Radius = behavioral STRANGENESS, not nonlinearity (nl peaks in shell 3, drops after). The cloud is a DISC (loops 5.1/3.8/2.0), and every data domain reshapes it - MNIST collapses it to a flat cigar (6.8/1.7/1.4). Baseline map per domain is meaningful.
3. Lens diversity alone makes hard targets linear: ridge on the bank hits $R^2 = 1.000$ where raw-linear scores ~ 0 . Frontier of the pointwise bank: $\sin 32x$ (frequency, not temporality).
4. Rotation proves the geometry carries information: co-planar lenses are REDUNDANT views; worst slices score far below random (0.17 vs 0.996 on loops); the good angle (~ 81 deg) is stable across bank sizes. Spin axis is irrelevant; whitening does not help.
5. Across domains, the less pointwise diversity buys (MNIST +3.9 pts, CIFAR +0.0, audio -3.8), the more the right SLICE beats the whole bank - on CIFAR 24 lenses at 105 deg (0.4015) beat both the 400-lens bank (0.3815) and raw pixels (0.3820).

What the space costs

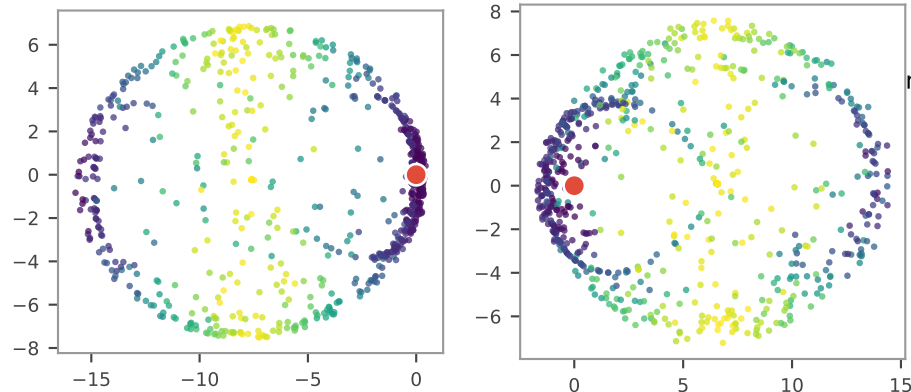
The space is generated, never stored: ~ 2 KB of code, a bank is a list of integers.
Model size = the linear head only: loops 1.6 KB, text-at-bigram-ceiling 26 KB, MNIST ~ 1 MB ($L=32$). The whole multi-domain baseline family fits in ~ 5 -15 MB.

Where genomes must begin (measured gaps)

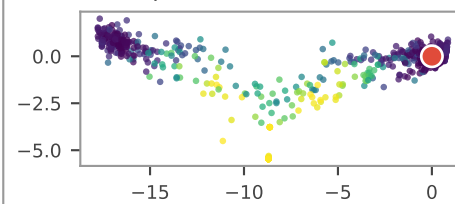
Pixel interactions (images) - context beyond the bigram ceiling 0.2661 (text) -
phase-invariant temporal structure (audio, where the pointwise bank actively hurts) -
and slice selection, already worth +2 pts on CIFAR with 24 lenses.

The baseline maps — one behavioral map per data domain (colour = nonlinearity, red dot = identity at the origin)

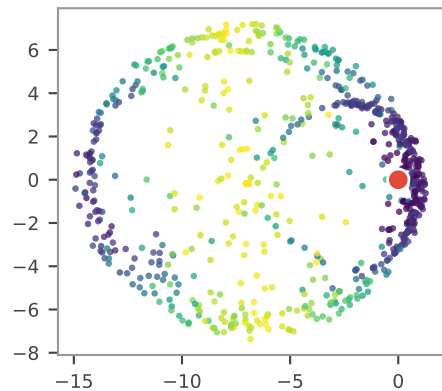
loops — shape 5.1 / 3.7 / 2.0 (r-vs-nl 0.482) noise — shape 4.8 / 3.7 / 2.1 (r-vs-nl 0.515)



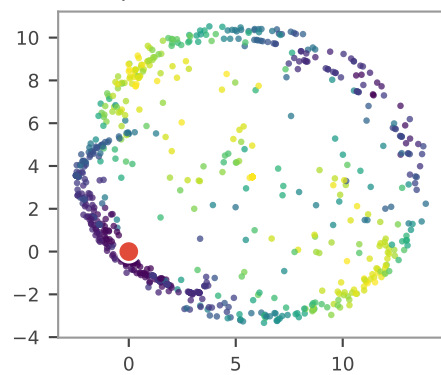
mnist — shape 6.8 / 1.7 / 1.4 (r-vs-nl 0.276)



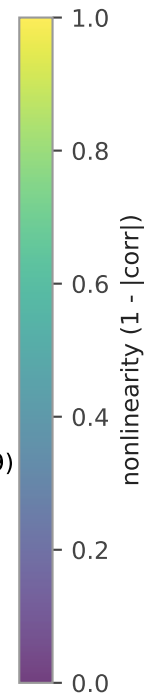
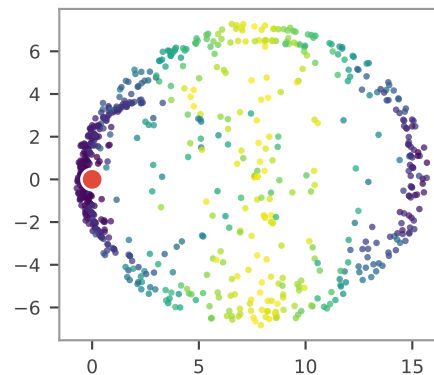
cifar — shape 4.8 / 3.7 / 2.1 (r-vs-nl 0.503)



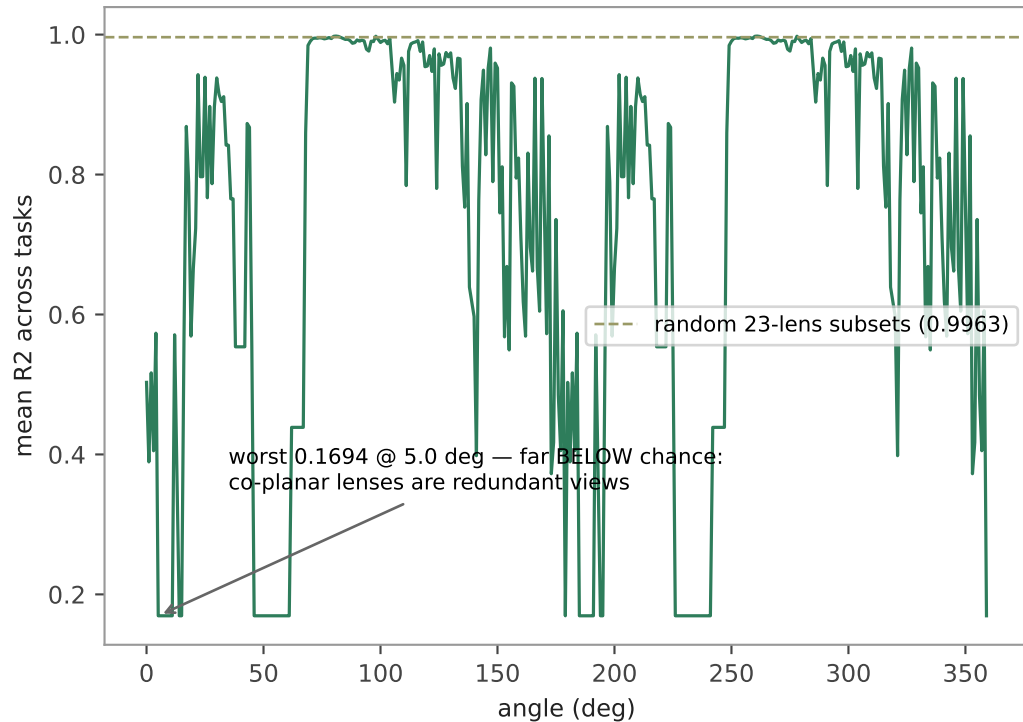
text — shape 4.8 / 4.0 / 2.1 (r-vs-nl 0.498)



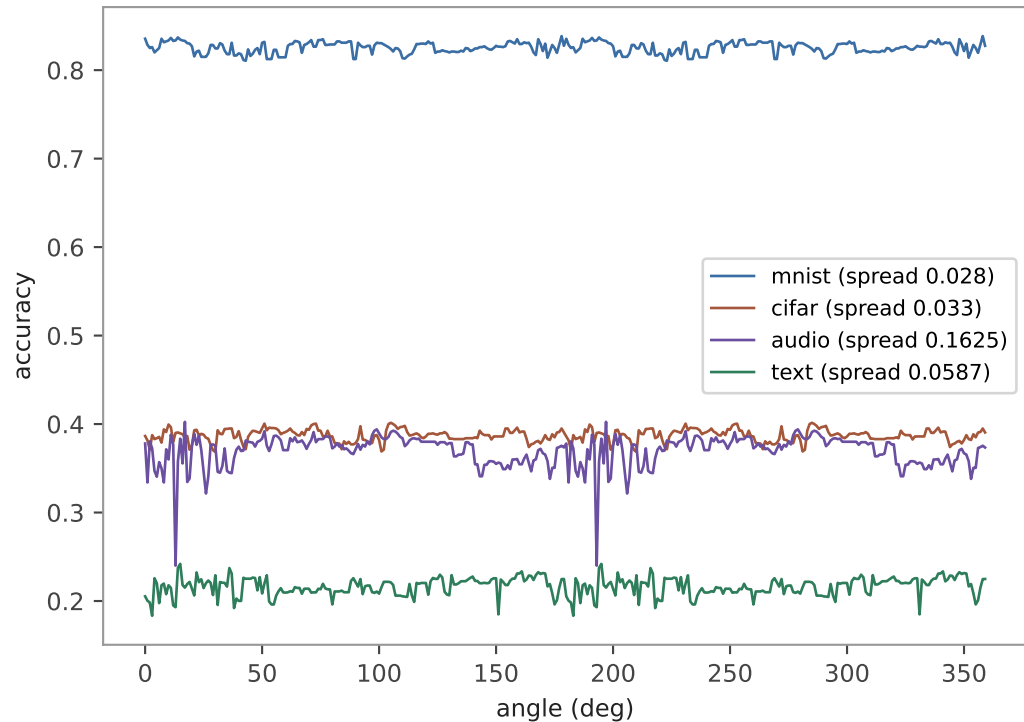
caltech — shape 5.1 / 3.5 / 1.9 (r-vs-nl 0.479)



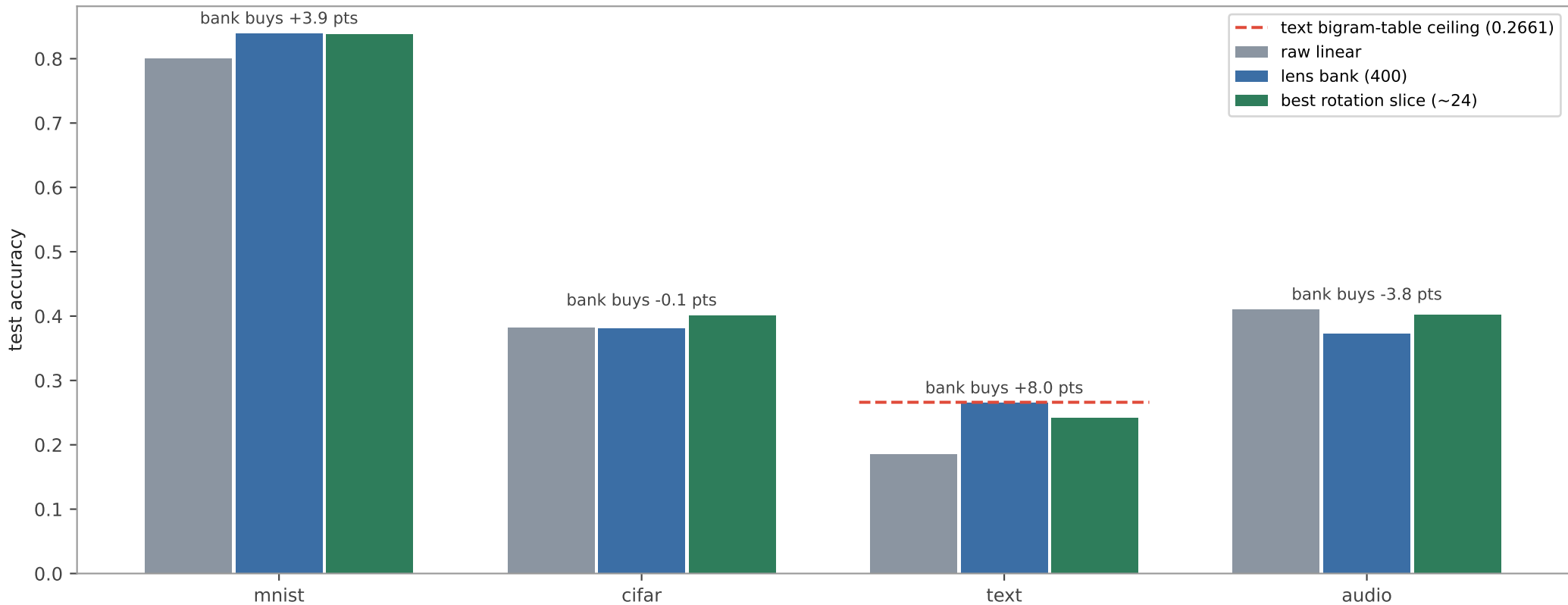
Loops: R2 vs rotation angle (slice ~23 of 748)



Domains: accuracy vs rotation angle (per-domain maps)



Domain baselines — pointwise lens bank + closed-form linear head, no genomes



Details, corrections and the measured gaps

Sources: radial_data/baseline_{mnist,cifar,text,audio}.json, prebaseline_fixes.json, CHANGELOG.md 2026-07-13 entries, doc

The map's structure (2500-lens loops map)

Near-linear lenses ($nl < 0.1$): radius 2.9 vs population 7.3 - the linear core is real.
Family ordering by mean radius: id/logp/soft/tanh (4.3-5.2) -> sign/step/relu (5.3-5.7)
-> sin (7.2) -> sq/abs/cos/gauss/sqrt/sinc (9.2-10.3). Oscillator decile forms a lobe.
CORRECTION: outermost shell $nl\ 0.43 < \text{shell-3}\ nl\ 0.88$ - distance encodes the whole behavioral signature (dead zones, tails, saturation), not nonlinearity alone.

Task ladder (auto-advance at $R2 \geq 0.998$, pointwise rungs only)

loops: square, abs, sin3x, ripple, sin8x, sin16x pass; FRONTIER sin32x (0.990).
noise: frontier sin16x (0.949). The wall is frequency resolution and it is data-dependent.

Rotation probe facts

Loops: best 81 deg 0.998 / worst 5 deg 0.169 / random 0.996 - worst is far below chance.
Best angle stable at ~81-82 deg for 800 and 2500 lenses. Axis lock: x/y/z spins all give spread ~0.83 and the SAME floor (0.169). Whitened axes: no change. Free rotation is fine.

Domain baselines (8k train / 2k test; text 60k/15k)

MNIST raw 0.8005 -> bank 0.8395 (saturates by $L=8-32$: compounding problem, not coverage).
Top lenses are gauss/abs folds (ink detectors). Hardest classes 5 (0.715), 2 (0.749).
CIFAR raw 0.3820 -> bank 0.3815 (zero). Best 24-lens slice 0.4015 beats everything.
Per-channel vs flat: identical under pointwise lenses (the split is invisible).
TEXT majority 0.1633, raw 0.1855, bank 0.2651 vs measured bigram ceiling 0.2661 -
within 0.001 with NO table; saturates exactly AT the ceiling. 8 lenses -> 0.2245.
AUDIO raw 0.4105 -> bank 0.3725 (the bank HURTS; random phase defeats pointwise views).
Largest angular spread (0.1625); best slice 0.4025 beats the bank.

Model sizes (the space itself is ~2 KB of generator code + integer addresses)

loops regressor 1.6 KB - text at the bigram ceiling ~26 KB (the table it replaces: 10 KB)
MNIST ~1 MB at $L=32$ (0.833), 12.5 MB at $L=400$ (0.8395) - CIFAR rational model = the
24-lens slice, ~3 MB. Parameters live in the linear head, not the representation.

Where genome evolution starts (the roadmap's closing line, now quantified)

1. Pixel interactions - the pointwise GAM ceiling is +3.9 pts (MNIST) / +0.0 (CIFAR).
2. Context beyond bigram - everything above 0.2661 on text needs more than one char.
3. Phase-invariant temporal structure - audio is where temporal rotation is mandatory.
4. Slice selection - an evolved slice-picker has measured headroom (+2.0 pts on CIFAR with 24 lenses); on hard domains WHICH lenses matters more than HOW MANY.

Round 2 — the four open questions, answered

Experiments of 2026-07-13 round 2; sources: radial_map.py analyses, radial_slice_ga.py, radial_data/slice_ga_cifar.json

Q1 - Why is the map a disc?

Axis 1 correlates +0.99 with LINEARITY - the dominant axis IS the linear-nonlinear direction (monotonic frac +0.60 rides along). Axis 2 is response-curve SHAPE (79% of curve variance in 2 PCs; no scalar stat explains it). Axis 3 is a weak oscillation mix. Lens behavior has ~2 dominant degrees of freedom: how linear, and what shape of bend.

Q2 - What is the dead-zone clump? (and is it removable?)

IT IS THE LINEAR CORE. Worst slice = 23 lenses, all outer-op id, nonlinearity 0.000, intra-clump signature distance 0.34 vs 9.10 random: twenty-three copies of x. Hence the floor 0.1694 = raw-linear performance exactly, and hence the SAME floor on every spin axis - every great circle passes through the origin. REMOVABLE: excluding the 126 near-linear lenses raises the floor 0.169 -> 0.822 and collapses spread 0.83 -> 0.18. Practical rule: never build a slice out of the core.

Q3 - Does temporal rotation work? CONFIRMED

Rotate a lens's view through time: features = agreement between the view at t and t+dt (dt = 1..16) - lens-space autocorrelation, deterministic, closed-form, phase-invariant by construction. Audio tones: raw 0.4105, pointwise bank 0.3725 -> TEMPORAL ROTATION 1.0000 with 16 features. Caveat, honestly held: the identity lens alone saturates this task, so the mechanism is proven while lens DIVERSITY in the temporal regime awaits a harder task.

Q4 - The genome bridge

First radial genome built and run (radial_slice_ga.py, batched GPU fitness: one (POP,N,N) LU per generation, 305 s total): a 24-lens slice picker on CIFAR. RESULT: 0.376 - random-subset level (0.3774), well below the map-geometry slice (0.4015). Fitness was flat by NATURE (random 24-subsets differ by +/-0.002), the classic flat-landscape signature (soft -2.074 -> -2.058, starved 0). Honest conclusion: naive evolution over raw lens indices cannot see the structure the map exposes for free. THE GENOME'S SEARCH SPACE SHOULD BE THE MAP ITSELF (plane / angle / region parameters, not indices) - the radial thesis restated as an evolution recipe, and the design brief for genome #2.